Improved efficiency for likelihood-based random effects meta-analysis

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Abstract. We consider Bartlett adjustment to improve the efficiency for the random effects meta-analysis based on the Gaussian likelihood ratio. We first derive the Bartlett correction for the idealized Gaussian case, correcting the version appearing in the literature. As the Gaussian assumption is restrictive, we give more general conditions under which the Bartlett rate of $O(1/n^2)$ can be achieved. In contrast with finite-sample normality the proposed condition involves asymptotic normality and a requirement on the order of the primary study sizes relative to the number of primary studies.

1 Background

Meta-analysis is a technique for synthesizing a body of related scientific results into a summary conclusion. Suppose each of n studies, all measuring the same underlying phenomenon, produces an estimate Y_i and standard error $\hat{\sigma}_i$. In the likelihood ratio approach to meta-analysis (Hardy and Thompson, 1996), the data are assumed to be independent and conditionally Gaussian,

$$Y_i \mid \hat{\sigma}_i^2 \sim N(\mu_0, \tau_0^2 + \hat{\sigma}_i^2), \qquad \tau_0^2 > 0 \tag{1}$$

That is, conditionally on $\hat{\sigma}_1^2, \ldots, \hat{\sigma}_n^2$, $Y_1, \ldots, Y_n$ have log-likelihood

$$l(\mu, \tau^2) = -\frac{1}{2}\sum_{i=1}^n \log(2\pi(\hat{\sigma}_i^2 + \tau^2)) - \frac{1}{2}\sum_{i=1}^n \frac{(Y_i - \mu)^2}{\hat{\sigma}_i^2 + \tau^2}.$$

The likelihood ratio ("LR") statistic for testing $\mu = \mu_0$ is

$$W = W(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE}, \hat{\tau}^2_{MLE,R}) = 2(l(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE}) - l(\mu_0, \hat{\tau}^2_{MLE,R})) \tag{2}$$

where $(\hat{\mu}_{MLE}, \hat{\tau}^2_{MLE})$ maximizes $l(\mu, \tau^2)$ and $\hat{\tau}^2_{MLE,R}$ maximizes $\tau^2 \mapsto l(\mu_0, \tau^2)$.

(Hardy and Thompson, 1996) proposed using the LR statistic so that the variance component τ^2 would be estimated simulatenously with μ, whereas previous methods did not take into account estimation error of the former. The approach also has the benefit of converging at a $1/n$ rate. See Michael (2026) for relatively general conditions under which the LR statistic attains a $1/n$ rate.

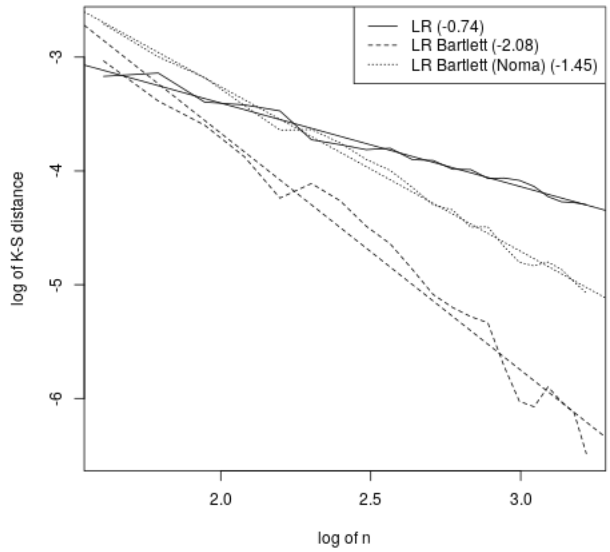


Figure 1: Bartlett correction in the Gaussian model (1). The figure is a log-log plot of the Kolmogorov-Smirnov distance between the observed test statistic and the chi-squared distribution versus the number of studies $n = 5, \ldots, 25$. The slopes of the fitted lines given in parentheses approximate the convergence rate exponent. The unadjustd LR is approximately $O(1/n)$, becoming $O(1/n^2)$ after Bartlett correction. The incomplete adjustment of Noma (2011) leads to a rate somewhere in between.

A challenge in carrying out meta-analysis is that the number of available studies is often small (Davey et al., 2011). Efficiency of the statistical procedures is therefore especially valuable. For inference based on the LR, Bartlett correction (Bartlett, 1937; Lawley, 1956) offers the tantalizing prospect of improving convergence rates from $1/n$ to $1/n^2$.

Proposition 1. *Suppose model* (1) *holds. Then the Bartlett correction for the LR statistic* (2) *is given by*

$$c = \frac{2\sum_{i=1}^n w_i^3}{\sum_{i=1}^n w_i \sum_{i=1}^n w_i^2} - \frac{(\sum_{i=1}^n w_i^2)^2}{2\sum_{i=1}^n w_i^2 (\sum_{i=1}^n w_i)^2}, \qquad \textit{where } w_i = (\hat{\sigma}_i^2 + \hat{\tau}_{MLE,R}^2)^{-1},$$

That is, under model (1)*,* $W/(1+c)$ *converges to a chi-squared distribution at an* $O(1/n^2)$ *rate (Kolmogorov-Smirnov distance).*

See Figure 1 for an illustration. This formula for the Bartlett correction corrects that given in Noma (2011), which omits the second of the two terms, ultimately due to a typo present in Cordeiro (1993). See the proof of Prop. 1 for details.

At first blush Bartlett correction for meta-analysis may appear incoherent. The Gaussian model for meta-analysis is understood to be a "working model." It may be a reasonable way to carry out inference on a location parameter under the constraint of a small sample size, but the model is directly contradicted by well-known properties of the data. Common choices of Y_i like the log-odds are not centered at the null μ_0, and they are correlated with their standard errors, unlike Gaussian data. Yet whereas this basic moment structure is already misspecified, the Bartlett correction presumes still deeper fidelity to the Gaussian distribution. The attempt appears to take too literally something that was only ever intended in a figurative sense.

On the other hand the Gaussian model does have a structural basis. It is based on the obesrvation that the primary studies are usually large, at least relative to the number of studies. That is, Y_i are asymptotically normal in the primary study sizes m_i. This observation is supported by the data. The common effect estimators and their variance estimators such as the log-odds or standardized mean difference are typically asymptotically normal as $m_i \to \infty$, at least conditionally given the random effect.

Suppose we pursue this approach naively. The effects Y_i generally converge to asymptotically normal variables Z_i, say, at a $1/\sqrt{m_i}$ rate. For simplicity suppose also that a meta-analysis consists only of single-arm studies all with the same sample size $m_i = m$. Schematically,

$$W\Big|_{\{Y_i\}_{i=1}^n \to \{Z_i\}_{i=1}^n} - W = O_P(1/\sqrt{m}) \cdot \frac{d}{d\{Y_i\}_{i=1}^n} W + \text{ higher order terms.}$$

The derivative of the LR statistic above is $O(\sqrt{n})$ when the null $\mathbb{E}Y_i = \mu_0$ is exactly or even approximately true (Michael, 2026). So the resulting misspecification error due to non-Gausian Y_i is generally $O(\sqrt{n})O(\frac{1}{\sqrt{m}})$. For the $1/n^2$ Bartlett rate to be achieved this heuristic argument would then suggest $O(\sqrt{n})O(\frac{1}{\sqrt{m}})$ must be $O(1/n^2)$, or that the typical study size m_i be on the order of n^5, which is unrealistic. Below we give relatively general conditions under which the $1/n^2$ rate may be achieved when m_i is $O(n^{3/2})$.

2 Main result

We obtain an asymptotic expansion of a Bartlett-corrected LR statistic that enables us to see when the $1/n^2$ Bartlett rate can be achieved. The proof proceeds by relating the heteroscedastic Gaussian LR statistic (2) to the simple homoscedsatic Gaussian LR statistic $W_0 = n \log(1 + (\frac{\overline{Y} - \mu_0}{S})^2)$. The latter is forgiving of misspecification since, to first order, it is a self-normalized mean, which is within $O_P(1/\sqrt{n})$ of a Gaussian random variable under general settings when the null is at least approximately true (Giné et al., 1997). On top of this asymptotic normality, the Y_i that make up the studentized statistic are often themselves within $O(1/\sqrt{m_i})$ of a Gaussian distribution. In several places in the proofs these two asymptotically normal convergence rates are found to multiply, enabling us to reduce the heuristic $m = O(n^5)$ condition to $m = O(n^{3/2})$.

Our asymptotic regime involves both the number of studies and the sizes of the studies. We quantify the latter using the variance estimates $\hat{\sigma}_i^2$, which we assume to be consistent and which vanish as the study sizes grow. We view $\hat{\sigma}_i^2$ as functions of the number of studies n, say $\hat{\sigma}_i^2(n)$, and arrange them as rows of a triangular array growing with n. We reduce these n quantities to a single quantity by stipulating that they are all of the same order, in the sense that there is an increasing sequence $m = m(n)$ such that

$$0 < \lim_{n\to\infty} m(n) \min_{1\le j\le n} \hat{\sigma}_i^2 \le \lim_{n\to\infty} m(n) \max_{1\le j\le n} \hat{\sigma}_i^2 < \infty. \tag{3}$$

This assumption allows us to assert, for example, that when the Y_i are asymptotically normal, they all converge at the same $1/\sqrt{m}$ rate. See Michael (2026) for further discussion.

Proposition 2 expresses the gap between the Bartlett-corrected LR statistic and its limiting chi-squared distribution in terms of the order of magnitude m of the studies and the average skew and kurtosis of the study effects Y_i.

Proposition 2. *Assume:*

1. *assumption* (3)
2. *assumptions of Lemma 4*
3. *assumptions of Lemma 5*

Let γ and κ denote the average skew and average excess kurtosis of $Y_1, \ldots, Y_n$. Then

$$\mathbb{P}(\frac{W}{1+3/(2n)} < x) = F_{\chi^2}(x) + \frac{1}{n}O(|\kappa| + \gamma^2) + \frac{1}{m}O(|\gamma|) + O_P(\frac{n}{m^2}) + O_P(\frac{1}{m\sqrt{n}}) + O_P(\frac{1}{n^2}).$$

The proposition does not require the data to be identically distributed, allows for correlation between the effects and variance estimates, and allows for finite-sample biases of the effect estimators, which are common features of meta-analysis data; see Section 1. The proposition does not itself invoke a random effect or latent variable. The main conditions are in the nature of regularity assumptions, such as requiring τ^2 be strictly positive, and the Cramér condition, excluding lattice Y_i.

The correction used in Proposition 2, $1+3/(2n)$, is the homoscedastic Gaussian correction (Lawley, 1956; McCullagh, 1987). It is the limit of the correction c given in Prop. 1 as $\hat{\sigma}_i^2 \to 0$, $i = 1, \ldots, n$, i.e., as the heteroscedastic Gaussian model turns into the homoscedastic Gaussian model. By contrast, the version of the correction proposed in Noma (2011) would imply a correction of $2/n$ for the homoscedastic Gaussian model. Under the conditions of Corollary 3 the simple homoscedastic correction is enough to eliminate approximation error up to $O(1/n^2)$.

Corollary 3. *Assume:*

1. *assumptions of Prop. 2*
2. $m = O(n^{3/2})$
3. *average standardized skew of Y_i is $O(1/\sqrt{m})$*
4. *average standardized kurtosis of Y_i is $O(1/m)$*

Let $c_0 = 3/(2n)$. Then the Bartlett-corrected LR statistic $W/(1 + c_0)$ converges to a chi-squared distribution at an $O(1/n^2)$ rate (Kolmogorov-Smirnov distance).

The required rates for the skew and kurtosis are met by most asymptotically normal quantities based on an average of independent random variables. In particular, the conditions hold when Y_i is one of the common effect estimators such as the log-odds ratio or SMD, and the actual effect in study i is obtained by a random draw from a normal distribution. This model generalizes the model (1) which requires not only that the random effect be Gaussian, but the study-level data. The asymptotic normality of the study-level data is sufficient. On the other hand, when the random effect has a non-Gaussian distribution, the convergence rate of the Bartlett-corrected LR statistic depends on that random effect's symmetry and excess kurtosis. If those cumulants are nonzero, the marginal distribution of Y_i may also have non-zero skew or kurtosis, blocking the $O(1/n^2)$ rate from being achieved. See Section 3 for a simulation illustrating the effect of skew and kurtosis on the rate.

3 Simulation

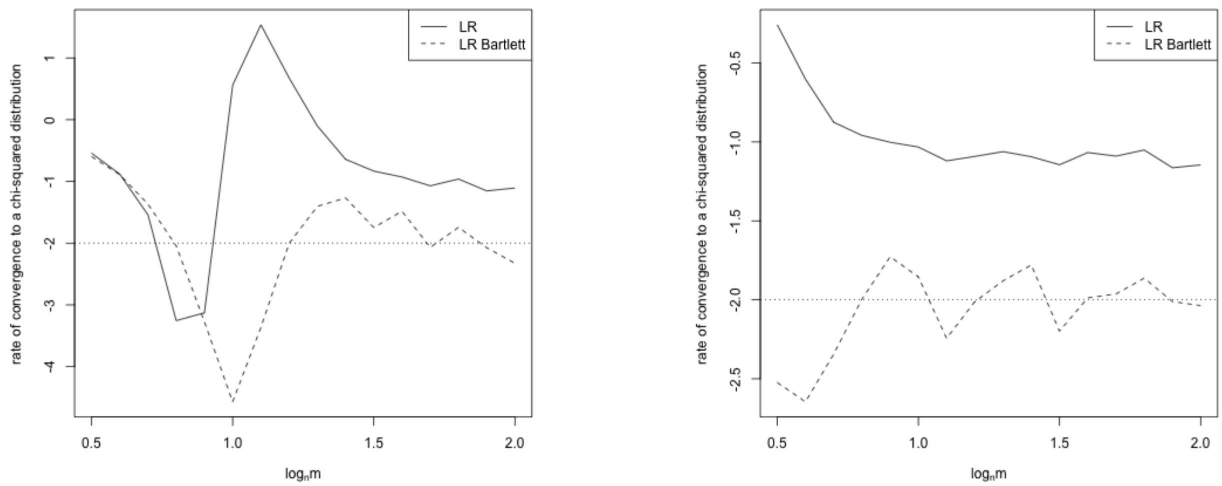
We examine by simulation the two main types of conditions imposed by Proposition 2 and Corollary 3 to achieve the $O(1/n^2)$ rate. The first is that the study sizes m be large enough, and the second is the approximate normality of Y_i.

Data is generated from hypothetical two-arm trials. The mean sample sizes across arms is imbalanced in the ratio 3 : 2. Given these mean sample sizes, the actual sample sizes m_{Ti} and m_{Ci} are sampled as Poisson, floored at 2. For each of n studies, a random effect B_i, representing the treatment effect underlying study i, is drawn with mean μ_0 and variance τ_0^2. In Section 3.1 B_i are Gaussian, while in Section 3.2 perturbed Gaussians are examined. We consider both discrete and continuous study-level data. For the discrete data, a control group event probability is sampled as beta with mean .35 . Cell probabilities are then sampled as binomial using the control group event probability, the treatment group event probability implied by the random effect, and the sample sizes m_{Ti} and m_{Ci}. Y_i is then computed as the log-odds ratio. For continuous data, the individual patient data is sampled as standard exponential, and Y_i is computed as the standardized mean difference.

All simulations are based on n^4 monte carlo repetitions, implying a monte carlo error of $O(1/n^2)$, which is needed for any hope at detecting $O(1/n^2)$ convergence. The sample sizes are $n = 5, \ldots, 20$ studies. The rate of convergence of an asymptotically chi-squared statistic is estimated as the slope coefficient in a log-log regression of the observed Kolmogorov-Smirnov distance against the number of studies n.

3.1 Study size assumption $m \geq n^{3/2}$

This simulation examines the impact of $\log_n m$ on the rate of convergence to 0 of the Komogorov-Smirnov distance between the Bartlett-corrected LR statistic and a chi-squared distribution. Figure 2 presents results for log-odds and SMDs. In both cases the $O(1/n^2)$ rate is achieved once $m \geq n^{3/2}$. In the case of the log-odds, however, there is a sharp drop in the rate from $O(1/n)$ to $O(1/n^2)$ in the vicinity of $\log_n m = 3/2$. The SMD achieves the $O(1/n^2)$ rate across the tested values of $\log_n m$. The uncorrected LR statistic is presented for comparison. In both cases it stabilizes at an $O(1/n)$ rate, as expected.



(a) Count data with the log-odds estimator.

(b) Exponential data with the SMD estimator.

Figure 2: The rate of convergence to a chi-squared distribution of the LR statistic with and without Bartlett correction, as m varies betwen $n^{1/2}$ to n^2. The target rate of $1/n^2$ is approximated once $m \geq n^{3/2}$ for the count data, as anticipated by Prop. 2, but for all m values for the continuous data, suggesting the conditions are sufficient but not necessary.

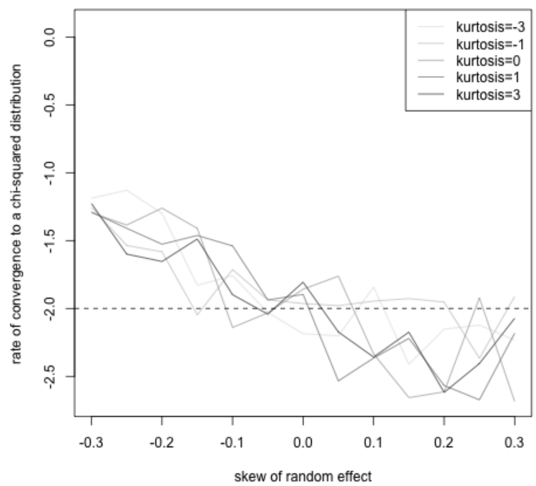
3.2 Normality of Y_i

We next consider how the convergence rate of the Bartlett-corrected LR statistic is affected when the random effect distribution is perturbed away from the Gaussian distribution. The perturbations are carried out by sampling B_i from the sinh-arcsinh family described by Jones and Pewsey (2009)

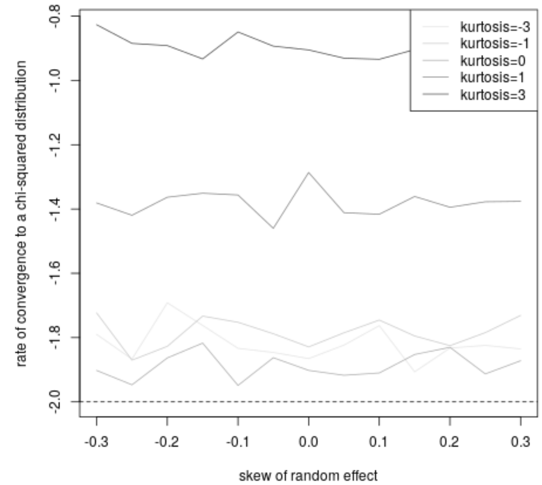
$$B_i = \sinh(\frac{\text{arcsinh}(Z_i) + \epsilon}{\delta}), \qquad Z_i \sim N(0,1).$$

In this two-parameter family, ϵ controls skew and δ controls kurtosis, with values of 0 and 1 giving a standard Gaussian. In this simulation the standardized skew of the random effect is varied between -.3 and .3, and the standardized excess kurtosis between -3 and 3. The order of the primary study sizes is fixed at $m = n^{3/2}$.

Results are presented in Figure 3. As expected, when the skew and excess kurtosis vanish, the corrected LR statistic approximates the $O(1/n^2)$ rate whether log-odds or SMD is used. The simulation does, however, reveal some surprises. First, whereas the log-odds ratio is sensitive to skew and apparently insensitive to non-zero excess kurtosis, the SMD simulations shows the opposite pattern. Second, one might expect that positive or negative skew should diminish the log-odds rate in a symmetric way. However, it appears that negative skew adversely affects the log-odds rate while positive skew boosts it, with rates exceeding the Bartlett-corrected $O(1/n^2)$ rate appearing at higher positive skew values.



(a) Count data with the log-odds estimator.



(b) Exponential data with the SMD estimator.

Figure 3: The rate of convergence to a chi-squared distribution of the LR statistic with and without Bartlett correction, as the skew and kurtosis of the random effect is varied. Whereas the count data appears sensitive to skew but not kurtosis, it appears the opposite holds for the continuous data.

4 Data analysis

We follow Noma (2011) in re-analyzing the meta-analysis data set of Teo et al. (1991), confirming the conclusions of (Noma, 2011). The scientific question is the effect of magnesium on mortality due to acute myocardial infarction. The initial analysis of Teo et al. (1991), using a fixed effects estimator, suggested a protective odds ratio of .47 with 95% CI (.28, .79), whereas a subsequent large randomized control trial (ISlS-4 Collaborative Group, 1995) resulted in a null or harmful finding of $1.06\ (1.00, 1.12)$.

See Table 1 for summary statistics of the primary studies. The average sample sizes in the control and treatment groups are $n_C = 92$ and $n_T = 94$, easily exceeding the $n^{3/2} = 19$ cutoff of Prop. 2. The MLE for the effect of the intervenion is .449 on the odds ratio scale. Table 1 presents CIs obtained using the LR statistic unadjusted by any correction factors, adjusted by the Bartlett correction of Noma (2011), adjusted by the Bartlett correction given by Prop. 1, and adjusted by the simplified homoscedastic Bartlett correction of Prop. 2. All adjusted rates return insificant results, though the simplified homoscedastic Bartlett correction is just on the boundary of significance. The CI of Noma (2011) and the repaired version of Prop. 1 are quite close for this data set, the latter being about 5% smaller.

Whereas publication bias is often cited a the culprit for the spurious findings of smaller meta-analyses, Teo et al. (1991) is an example where a more accurate statistical procedure would have avoided a misleading conclusion.

Study	Trt	Control	Log-OR	SE
Morton et al.	1/40	2/36	−0.83	1.25
Rasmussen et al.	9/135	23/135	−1.06	0.41
Smith et al.	2/200	7/200	−1.27	0.81
Abraham et al.	1/48	1/46	−0.04	1.43
Feldstedt et al.	10/150	8/148	0.22	0.49
Schechter et al.	1/59	9/56	−2.41	1.07
Ceremuzynski et al.	1/25	3/23	−1.28	1.19

test	95% CI
LR	(0.192, 0.903)
LR, Bartlett	(0.165, 1.030)
LR, Bartlett (Noma)	(0.158, 1.066)
LR, Bartlett (simple)	(0.170, 0.992)

Table 1: Summary statistics for the meta-analysis given in Teo et al. (1991) and 95% CIs of LR tests.

5 Discussion

When little data is available, achieving good rates naturally requires strong modelling assumptions. For Noma (2011) these assumptions are the "normal-normal" model, where the study effects Y_i are obtained by convolving a Gaussian random effect with a Gaussian study-level effect. One perspective on what we have done is to replace the "normal-normal" model with a "normal" model, instead requiring an approximately Gaussian random effect and a lower bound on m. While the normal-normal model is directly contradicted by typical meta-analysis data, a Gaussian random effect is a consistent, if unidentifiable, postulate. The latter may be subject to a sensitivity analysis, for example.

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Proof of Proposition 1. As verified by Noma (2011), the parameters μ and τ^2 are orthogonal under the conditional Gaussian model (1). Therefore the simplified matrix approach to Bartlett correction given in Cordeiro (1993) applies. The intended formula for the correction, which Noma (2011) relies on, is given by equation (10) of Cordeiro (1993). The formula, as written there, is

$$\begin{aligned} c = 1 &+ k_{\alpha,\alpha}^{-1}(\mathbb{1}^T M \mathbb{1} + a^T K_{\beta\beta}^{-1} d + \frac{1}{4}(\mathbb{1}^T (K_{\beta\beta}^{-1} * B)\mathbb{1})^2) \\ &+ k_{\alpha,\alpha}^{-2}(\alpha_1 + a^T K_{\beta\beta}^{-1} d + \alpha_2 \mathbb{1}^T (K_{\beta\beta}^{-1} * B)\mathbb{1}) + k_{\alpha,\alpha}^{-3}\alpha_3. \end{aligned} \tag{4}$$

Several transcription/typographical errors appear upon comparing equation (10) of Cordeiro (1993) with the displays immediately preceding it. Repairing these, the formula becomes

$$\begin{aligned} c = 1 &+ k_{\alpha,\alpha}^{-1}(\mathbb{1}^T M \mathbb{1} + a^T K_{\beta\beta}^{-1} d + \frac{1}{4}(\mathbb{1}^T (K_{\beta\beta}^{-1} * B)\mathbb{1})^2) \\ &+ k_{\alpha,\alpha}^{-2}(\alpha_1 + a^T K_{\beta\beta}^{-1} b + \alpha_3 \mathbb{1}^T (K_{\beta\beta}^{-1} * B)\mathbb{1}) + k_{\alpha,\alpha}^{-3}\alpha_2. \end{aligned}$$

The material typo is $b \to d$ in the term $a^T K_{\beta\beta}^{-1} b$. As Noma (2011) shows, $d = 0$ in the Gaussian model, so that term does not appear in the formula for the correction if (4) is used. When one substitutes

$$b = \frac{3}{4}\mathbb{E}(\frac{d^3 l(\mu,\tau^2)}{d\mu d\mu d(\tau^2)}) - \frac{d}{d\tau^2}\mathbb{E}(\frac{d^2 l(\mu,\tau^2)}{d\mu d\mu}) = -\frac{1}{4}\sum_{i=1}^{n} \frac{1}{(\tau^2+\sigma_i^2)^2},$$

keeping everything else from Noma (2011), one obtains the formula given in the proposition. $\square$

Notation: We use P_n for the empirical mean, i.e., sample average. We use E_n, var_n, and Cov_n to refer to average cumulants, e.g., $E_n(YV) = \frac{1}{n}\sum_{i=1}^n \mathbb{E}(Y_iV_i)$. We use $P_n - E_n$ to refer to the centered empirical average, e.g., $(P_n - E_n)(YV) = \frac{1}{n}\sum_{i=1}^n (Y_iV_i - \mathbb{E}(Y_iV_i))$. We use $\gamma =$ and κ to refer to the standardized average third and fourth cuwmulants of Y_i.

Proof of Proposition 2. Expand W in $1/m$ to get

$$\begin{aligned}
W &= n\log(1 + (\frac{T}{\sqrt{n}})^2) + \frac{1}{m}W' + O_P(\frac{n}{m^2}) \\
&= T^2 - \frac{T^4}{n} + O_P(\frac{1}{n^2}) + \frac{1}{m}W' + O_P(\frac{n}{m^2}) \\
&= (T - \frac{T^3}{4n} + O_P(\frac{1}{n^2}) + \frac{W'}{2m}(T - \frac{T^3}{4n} + O_P(\frac{1}{n^2}))^{-1})^2 + O_P(\frac{n}{m^2}) \\
&= R^2 + O_P(\frac{n}{m^2}) + O_P(\frac{1}{m\sqrt{n}})
\end{aligned}$$

where

$$\begin{aligned}
T &= \sqrt{n}\frac{P_n(Y) - \mu_0}{S} \\
R &= T - \frac{T^3}{4n} + \frac{W'}{2mT},
\end{aligned}$$

and the derivative

$$W' = \frac{T^2}{S^2S_0^4}(S^2 + S_0^2)n^{-1/2}\sum_i V_i((Y_i - P_n(Y))^2 - S^2) - \frac{2TS}{S_0^3}n^{-1/2}\sum_i(Y_i - P_n(Y)) \quad (5)$$

is $O_P(\sqrt{n})$ under the assumption $P_n(Y) - \mu_0 = O_P(\frac{1}{\sqrt{n}})$. R approximates the directed likelihood ratio statistic, which is typically asymptotically normal. Rather than attempting a direct Edgeworth expansion of the asymptotically chi-squared LR statistic, Lemmas 4 and 5 obtain an Edgeworth expansion for R, which is then used to expand the LR statistic.

Continuing, the CDF of W is

$$\begin{aligned}
\mathbb{P}(W < x) &= \mathbb{P}(R^2 < x) + O_P(\frac{n}{m^2}) + O_P(\frac{1}{m\sqrt{n}}) \\
&= \mathbb{P}(R < \sqrt{x}) - \mathbb{P}(R < -\sqrt{x}) + O_P(\frac{n}{m^2}) + O_P(\frac{1}{m\sqrt{n}})
\end{aligned}$$

Let $T_c = \sqrt{n}\frac{P_n(Y) - E_n(Y)}{S}$ and $b = E_n(Y) - \mu_0$, so that $T = T_c + \frac{\sqrt{n}}{S}b$. By Lemma 4,

$$\begin{aligned}
\mathbb{P}(W < x) = &\ \mathbb{P}(T_c - \frac{T_c^3}{4n} < \sqrt{x}) - \mathbb{P}(T_c - \frac{T_c^3}{4n} < -\sqrt{x}) \\
&+ (dF_{T_c}(\sqrt{x} + \frac{x^{3/2}}{4n}) - dF_{T_c}(-\sqrt{x} - \frac{x^{3/2}}{4n}))\left(\frac{\sqrt{n}}{\sigma}b(1 + O_P(\frac{1}{\sqrt{n}})) - \frac{\sqrt{n}}{m\sigma^3}E_n((Y - \mu_0)V)\right) \\
&+ (dF_{T_c}(\sqrt{x} + \frac{x^{3/2}}{4n}) + dF_{T_c}(-\sqrt{x} - \frac{x^{3/2}}{4n}))\frac{3\gamma}{2m\sigma^3}E_n((Y - \mu_0)V) + O_P(\frac{n}{m^2}) + O_P(\frac{1}{m\sqrt{n}}) + O_P(
\end{aligned}$$

The asymmetry of the density $dF_{T_c}(\sqrt{x}+\frac{x^{3/2}}{4n})-dF_{T_c}(-\sqrt{x}-\frac{x^{3/2}}{4n})$ vanishes with the skew $\mathrm{cum}(T_c,T_c,T_c)=\gamma\sigma^3/\sqrt{n}(1+O_P(1/\sqrt{n}))$, where γ is the standardized average third cumulant of $Y_1,\ldots,Y_n$. After this substitution,

$$\mathbb{P}(W<x)=\mathbb{P}(T_c-\frac{T_c^3}{4n}<r)-\mathbb{P}(T_c-\frac{T_c^3}{4n}<-r)+\frac{1}{m}O(|\gamma|)+O_P(\frac{n}{m^2})+O_P(\frac{1}{m\sqrt{n}})+O_P(\frac{1}{n^2}).$$

Let $F_{\chi^2}(x)=\Phi(\sqrt{x})-\Phi(-\sqrt{x})$ and $f_{\chi^2}(x)=\phi(\sqrt{x})/\sqrt{x}$ denote the CDF and PDF of a chi-squared random variable with 1 degree of freedom. By Lemma 5,

$$\mathbb{P}(W<x)=F_{\chi^2}(x)-\frac{\phi(r)}{n}\frac{3}{2}\sqrt{x}+\frac{1}{n}O(|\kappa|+\gamma^2)+\frac{1}{m}O(|\gamma|)+O_P(\frac{n}{m^2})+O_P(\frac{1}{m\sqrt{n}})+O_P(\frac{1}{n^2})$$

Applying the homoscedastic Gaussian Bartlett correction $3/(2n)$,

$$\begin{aligned}
\mathbb{P}(\frac{W}{1+3/(2n)}<x)&=\mathbb{P}(W<x(1+\frac{3}{2n}))\\
&=F_{\chi^2}(x)+\frac{3x}{2n}f_{\chi^2}(x)-\frac{\phi(\sqrt{x})}{n}\frac{3}{2}\sqrt{x(1+\frac{3}{2n})}+\frac{1}{n}O(|\kappa|+\gamma^2)+\frac{1}{m}O(|\gamma|)+O_P(\frac{n}{m^2})\\
&+O_P(\frac{1}{m\sqrt{n}})+O_P(\frac{1}{n^2})\\
&=F_{\chi^2}(x)+\frac{1}{n}O(|\kappa|+\gamma^2)+\frac{1}{m}O(|\gamma|)+O_P(\frac{n}{m^2})+O_P(\frac{1}{m\sqrt{n}})+O_P(\frac{1}{n^2})
\end{aligned}$$

□

Lemma 4 expresses the CDF of $R=T_c-\frac{T_c^3}{4n}+\frac{W'}{2mT}$ in terms of the CDF of $T_c-\frac{T_c^3}{4n}$, which is the homoscedastic Gaussian likelihood ratio statistic assuming the effects Y_i are unbiased for μ_0, and Lemma 5 gives the CDF of the latter.

Lemma 4. *Assume*

1. $P_n(Y)-\mu_0=O_P(1/\sqrt{n})$
2. $n^{-1/2}\sum_{i=1}^n(\mathbb{E}Y_i-\mu_0)^2=o_P(1)$
3. $\sup_i\mathbb{E}V_i^{2+\delta}<\infty$, $\sup_i\mathbb{E}Y_i^{4+\delta}<\infty$ *for some* $\delta>0$
4. $n=o(m^2)$
5. $b=O_P(1/m)$

Then

$$\mathbb{P}(R<x)=\mathbb{P}(T_c-\frac{T_c^3}{4n}<x)$$
$$+dF_{T_c}(x+\frac{x^3}{4n})\left(\frac{\sqrt{n}}{\sigma}b(1+O_P(\frac{1}{\sqrt{n}}))+\frac{1}{m}\frac{E_n((Y-\mu_0)V)}{\sigma^3}(\frac{3}{2}(x+\frac{x^3}{4n})\gamma-\sqrt{n})\right)$$
$$+O_P(\frac{1}{m\sqrt{n}})+O(\frac{1}{n^2})+O_P(\frac{n}{m^2})$$

where

$$b=E_n(Y)-\mu_0$$
$$\gamma=\frac{\frac{1}{n}\sum_{i=1}^n\mathbb{E}(Y_i-\mathbb{E}Y_i)^3}{(\frac{1}{n}\sum_{i=1}^n\mathbb{E}(Y_i-\mathbb{E}Y_i)^2)^{3/2}}.$$

Proof of Lemma 4.

$$R=T-\frac{T^3}{4n}+\frac{W'}{2mT}+O_P(\frac{1}{m\sqrt{n}})$$
$$=T_c-\frac{T_c^3}{4n}+\frac{\sqrt{n}}{S}b+\frac{W'}{2mT}+O_P(\frac{1}{m\sqrt{n}}).$$

Define $H=(\frac{\sqrt{n}}{S}b+\frac{W'}{2mT})/\epsilon$, $\epsilon=\frac{\sqrt{n}}{m}$, $b=E_n(Y)-\mu_0$, so that H is $O_P(1)$ when $b=O_P(1/m)$ and $W'=O_P(\sqrt{n})$. With this notation the CDF of R is

$$\mathbb{P}(R<x)=\mathbb{P}(T_c-\frac{T_c^3}{4n}<x-\epsilon H)$$
$$=\mathbb{P}(T_c-\frac{T_c^3}{4n}<x)+\epsilon\frac{d}{d\epsilon}\mathbb{E}\{T_c-\frac{T_c^3}{4n}<x-\epsilon H\}|_{\epsilon\to 0}+O_P(\epsilon^2). \quad (6)$$

The derivative is

$$\frac{d}{d\epsilon}\mathbb{E}\{T_c-\frac{T_c^3}{4n}<x-\epsilon H\}|_{\epsilon\to 0}=\mathbb{E}(\frac{d}{d\epsilon}\{T_c-\frac{T_c^3}{4n}<x-\epsilon H\})|_{\epsilon\to 0}$$
$$=\mathbb{E}(H\delta_{T_c-\frac{T_c^3}{4n}}(x-\epsilon H))|_{\epsilon\to 0}$$
$$=\int h\int\delta_{T_c-\frac{T_c^3}{4n}}(x-\epsilon H))dF_{T_c,H}(t,h)|_{\epsilon\to 0}.$$

By Lagrange inversion, $T_c-\frac{T_c^3}{4n}=x$ implies $T=x+\frac{x^3}{4n}+O(\frac{1}{n^2})$, so

$$\frac{d}{d\epsilon}\mathbb{E}\{T_c-\frac{T_c^3}{4n}<x-\epsilon H\}|_{\epsilon\to 0}=\int h\int\delta_{T_c}(x-\epsilon H+\frac{(x-\epsilon H)^3}{4n}+O(\frac{1}{n^2})))dF_{T_c,H}(t,h)|_{\epsilon\to 0}$$
$$=\int hdF_{T_c,H}(x+\frac{x^3}{4n},h)+O(\frac{1}{n^2})$$
$$=dF_{Tc}(x+\frac{x^3}{4n})\mathbb{E}(H\mid T_c=x+\frac{x^3}{4n})+O(\frac{1}{n^2}).$$

Substituting this expression back into (6),

$$\begin{aligned}\mathbb{P}(R < x) &= \mathbb{P}(T_c - \frac{T_c^3}{4n} < x) \\ &\quad + dF_T(x + \frac{x^3}{4n}) \left(\frac{\sqrt{n}}{\sigma} b(1 + O_P(\frac{1}{\sqrt{n}})) + \frac{1}{m}\mathbb{E}(\frac{W'}{2T} \mid T = x + \frac{x^3}{4n}) \right) + O(\frac{1}{n^2}) + O_P(\frac{n}{m^2}).\end{aligned}$$

We next simplify the conditional expectation, first finding an $O_P(1/\sqrt{n})$ approximation to the derivative W' and then using asymptotic bivariate normality to apply the familiar regression formulas.

The first of the two terms of that make up W' in (5) is

$$\begin{aligned}&P_n(Y - \mu_0)\frac{1}{\sqrt{n}}\sum_i V_i((Y_i - P_n(Y))^2 - S^2) \cdot \frac{S^2 + S_0^2}{2S^3 S_0^4} \\ &= P_n(Y - \mu_0)\frac{1}{\sqrt{n}}\sum_i V_i((Y_i - P_n(Y))^2 - S^2)\sigma^{-5} + O_P(\frac{1}{\sqrt{n}}) \\ &= \sqrt{n}P_n(Y - \mu_0)\sigma^{-5}(P_n(V(Y - \mu_0)^2) - S^2 P_n(V)) + O_P(\frac{1}{\sqrt{n}}) \\ &= T\sigma^{-4}(E_n(V(Y - \mu_0)^2) - \sigma^2 P_n(V)) + O_P(\frac{1}{\sqrt{n}}). \qquad (7)\end{aligned}$$

The second of the terms is

$$\begin{aligned}&\frac{S}{S_0^4}\sqrt{n}(P_n((Y - \mu_0)V) - P_n(V)P_n(Y - \mu_0)) \\ &= \frac{S}{S_0^4}(\sqrt{n}(P_n - E_n)((Y - \mu_0)V) + \sqrt{n}E_n((Y - \mu_0)V) - TSE_n(V)) + O_P(\frac{1}{\sqrt{n}}) \\ &= S^{-3}(\sqrt{n}((P_n - E_n)((Y - \mu_0)V) + \sqrt{n}E_n((Y - \mu_0)V) - TSE_n(V)) + O_P(\frac{1}{\sqrt{n}}) \\ &= (\sigma^{-3}(1 - \frac{3}{2}\frac{S^2 - \sigma^2}{\sigma^2} + O_P(\frac{1}{n})) \\ &\cdot ((\sqrt{n}(P_n - E_n)((Y - \mu_0)V) + \sqrt{n}E_n((Y - \mu_0)V) - TSE_n(V)) + O_P(\frac{1}{\sqrt{n}})) \\ &= \sigma^{-3}((1 - \frac{3}{2}\frac{S^2 - \sigma^2}{\sigma^2})\sqrt{n}E_n((Y - \mu_0)V) + \sqrt{n}(P_n - E_n)((Y - \mu_0)V) - TSE_n(V)) + O_P(\frac{1}{\sqrt{n}})). \qquad (8)\end{aligned}$$

Combining (7) and (8), a cancellation occurs as $T/\sigma^2 P_n(V) - T/\sigma^2 E_n(V) = O_P(1/\sqrt{n})$, leaving

$$\frac{W'}{2T} = \frac{T}{\sigma^2}E_n(V(Y-\mu_0)^2)$$
$$-\frac{1}{\sigma^3}\left((1-\frac{3}{2}\frac{S^2-\sigma^2}{\sigma^2})\sqrt{n}E_n((Y-\mu_0)V) + \sqrt{n}(P_n-E_n)((Y-\mu_0)V) - TSE_n(V)\right) + O_P(\frac{1}{\sqrt{n}})$$

$$= \frac{T_c}{\sigma^2}E_n(V(Y-\mu_0)^2) - \frac{1}{\sigma^3}\left((1-\frac{3}{2}\frac{S^2-\sigma^2}{\sigma^2})\sqrt{n}E_n((Y-\mu_0)V) + \sqrt{n}(P_n-E_n)((Y-\mu_0)V) - T_cSE_n(\right.$$

$$+\sqrt{n}b(\frac{E_n((Y-\mu_0)V)}{S} - \frac{E_n(V)}{\sigma^3}) + O_P(\frac{1}{\sqrt{n}}). \tag{9}$$

While $\sqrt{n}E_n((Y-\mu_0)V)$ is $O(\sqrt{n})$, the remaining terms on the RHS are $O_P(1)$.

The moment assumptions on Y imply the asymptotic joint normality of $(T_c, \sqrt{n}(P_n - E_n)((Y-E_n(Y))^2))$, in turn implying

$$\mathbb{E}(\sqrt{n}(S^2 - Var_n(Y)) \mid T_c = x + \frac{x^3}{4n}) = \sqrt{n}\mathbb{E}((P_n-E_n)((Y-E_n(Y))^2) \mid T_c = x+\frac{x^3}{4n}) + O_P(\frac{1}{\sqrt{n}})$$

$$= (x+\frac{x^3}{4n})/\sigma Cov_n((Y-E_n(Y))^2, Y-E_n(Y)) + O_P(\frac{1}{\sqrt{n}})$$

$$= (x+\frac{x^3}{4n})\sigma^2\gamma + O_P(\frac{1}{\sqrt{n}}). \tag{10}$$

The asymptotic joint normality of $(T_c, \sqrt{n}(P_n - E_n)(V(Y-\mu_0)))$ implies

$$\mathbb{E}(\sqrt{n}(P_n-E_n)(V(Y-\mu_0)) \mid T_c = x+\frac{x^3}{4n}) = (x+\frac{x^3}{4n})/\sigma Cov_n(V(Y-\mu_0), Y) + O_P(\frac{1}{\sqrt{n}})$$

$$= (x+\frac{x^3}{4n})/\sigma E_n(V(Y-\mu_0)^2) + O_P(\frac{1}{\sqrt{n}}). \tag{11}$$

Combining the two conditional expectations (10) and (11) into (9), $(x+\frac{x^3}{4n})\frac{1}{\sigma^4}E_n((Y -$

$\mu_0)^2V)$ adds out, leaving

$$\begin{aligned}
\mathbb{E}(\frac{W'}{2T} \mid T_c = x + \frac{x^3}{4n}) &= (x+\frac{x^3}{4n})\frac{1}{\sigma^4}E_n((Y-\mu_0)^2V) - \frac{1}{\sigma^3}\sqrt{n}E_n((Y-\mu_0)V) \\
&+ \frac{3}{2\sigma^5}E_n(V(Y-\mu_0))\mathbb{E}(\sqrt{n}(S^2-\sigma^2) \mid T_c = x + \frac{x^3}{4n}) \\
&- \frac{1}{\sigma^3}\mathbb{E}((P_n - E_n)(\sqrt{n}V(Y-\mu_0)) \mid T_c = x + \frac{x^3}{4n}) \\
&+ \sqrt{n}b(\frac{E_n((Y-\mu_0)V)}{S} - \frac{E_n(V)}{\sigma^3}) + O_P(\frac{1}{\sqrt{n}}) \\
&= -\frac{1}{\sigma^3}\sqrt{n}E_n((Y-\mu_0)V) + \frac{3}{2\sigma^3}\mathbb{E}(Y-\mu_0)V(x+\frac{x^3}{4n})\gamma \\
&+ \sqrt{n}b(\frac{E_n((Y-\mu_0)V)}{S} - \frac{E_n(V)}{\sigma^3}) + O_P(\frac{1}{\sqrt{n}}).
\end{aligned}$$

□

Lemma 5. *Assuming*

1. $0 < \lim_{n\to\infty} Var_n(Y) < \infty$
2. $\lim_{n\to\infty} \max_i \frac{VarY_i}{\sum_{j=1}^n VarY_j} \to 0$
3. *INID Cramér condition (Skovgaard, 1981)*
4. $\mathbb{E}|Y_i|^5 < \infty, i = 1,\ldots,n$

Then

$$\mathbb{P}(T_c - \frac{T_c^3}{4n} < x) - \mathbb{P}(T_c - \frac{T_c^3}{4n} < -x) = \Phi(x) - \Phi(-x) - \frac{\phi(x)}{n}\frac{3}{2}x + \frac{1}{n}O(\kappa+\gamma^2) + O(\frac{1}{n^2})$$

where

$$\begin{aligned}
\gamma &= \frac{\frac{1}{n}\sum_{i=1}^n \mathbb{E}(Y_i - \mathbb{E}Y_i)^3}{(\frac{1}{n}\sum_{i=1}^n \mathbb{E}(Y_i-\mathbb{E}Y_i)^2)^{3/2}} \\
\kappa &= \frac{\frac{1}{n}\sum_{i=1}^n \mathbb{E}(Y_i - \mathbb{E}Y_i)^4 - 3}{(\frac{1}{n}\sum_{i=1}^n \mathbb{E}(Y_i-\mathbb{E}Y_i)^2)^{2}}
\end{aligned}$$

Proof of Lemma 5. Under the given assumptions the studentized mean T_c has a valid Edgeworth expansion (Skovgaard, 1981)

$$\mathbb{P}(T_c < x) = \Phi(x) + \sum_{j=1}^3 n^{-j/2}p_j(x)\phi(x) + O(\frac{1}{n^2}),$$

where $p_j(x)$ are polynomials that may depend on the average cumulants/moments of Y_i. Furthermore, the parity of p_j alternates, with $p_1(x)$ being an even function, so that

$$\mathbb{P}(T_c < x) - \mathbb{P}(T_c < -x) = \Phi(x) - \Phi(-x) + \frac{\phi(x)}{n} 2p_2(x) + O(\frac{1}{n^2}).$$

Upon substituting the value for $p_2(x)$ computed in Hall (1992), where the cumulants are now interpreted as standardized average cumulants of the data,

$$\mathbb{P}(T_c < x) - \mathbb{P}(T_c < -x) = \Phi(x) - \Phi(-x) + \frac{\phi(x)}{n}((\frac{\kappa}{2} + \frac{\gamma^3}{3} - \frac{3}{2})x + (\frac{\kappa}{6} - \frac{2\gamma^2}{9} - \frac{1}{2})x^3 - \frac{1}{9}\gamma^2 x^5) + O(\frac{1}{n^2}).$$

Therefore

$$\begin{aligned}
\mathbb{P}(T_c - \frac{T_c^3}{4n} < x) - \mathbb{P}(T_c - \frac{T_c^3}{4n} < -x) &= \mathbb{P}(T_c < x + \frac{x^3}{4n}) - \mathbb{P}(T_c < -x - \frac{x^3}{4n}) + O(\frac{1}{n^2}) \\
&= \Phi(x) - \Phi(-x) + \frac{\phi(x)x^3}{2n} + \frac{\phi(x)}{n}((\frac{\kappa}{2} + \frac{\gamma^3}{3} - \frac{3}{2})x + (\frac{\kappa}{6} - \frac{2\gamma^2}{9} - \frac{1}{2})x^3 - \frac{1}{9}\gamma^2 x^5) + O(\frac{1}{n^2}) \\
&= \Phi(x) - \Phi(-x) + \frac{\phi(x)x^3}{2n} + \frac{\phi(x)}{n}((\frac{\kappa}{2} + \frac{\gamma^3}{3} - \frac{3}{2})x + (\frac{\kappa}{6} - \frac{2\gamma^2}{9})x^3 - \frac{1}{9}\gamma^2 x^5) + O(\frac{1}{n^2}) \\
&= \Phi(x) - \Phi(-x) - \frac{\phi(x)}{n}\frac{3}{2}x + \frac{1}{n}O(\kappa + \gamma^2) + O(\frac{1}{n^2}).
\end{aligned}$$

□